

4.1

(a) 信道转移矩阵:

$$P = \begin{bmatrix} 1-\varepsilon-\delta & \varepsilon & \delta \\ \delta & \varepsilon & 1-\varepsilon-\delta \end{bmatrix} \quad \therefore H(Y|X) = H(1-\varepsilon-\delta, \varepsilon, \delta)$$

设 $P(X=0) = p \quad \therefore P(X=1) = 1-p$

$\therefore P(Y=1) = p\varepsilon + (1-p)\delta = \varepsilon$ 始终是这个.

$\therefore$ 当 $P(Y=0) = P(Y=2) = \frac{1-\varepsilon}{2}$ 时最大

$\therefore H(Y)_{\max} = -2\left(\frac{1-\varepsilon}{2} \log_2 \frac{1-\varepsilon}{2}\right) - \varepsilon \log_2 \varepsilon$

$\therefore C = \max [H(Y) - H(Y|X)] = 1-\varepsilon - \varepsilon \log_2 \varepsilon - (1-\varepsilon) \log_2 (1-\varepsilon) - H(1-\varepsilon-\delta, \varepsilon, \delta)$

(b) $P = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 0.5 & 0.5 \\ 0 & 0.5 & 0.5 \end{bmatrix}$ 设输入概率为 $[p_0, p_1, p_2]$

$\therefore P(Y=0) = p_0 + 0.5p_1 \quad P(Y=1) = 0.5p_1 + 0.5p_2$
 $P(Y=2) = 0.5p_1 + 0.5p_2$

发现 $P(Y=1) = P(Y=2)$ 设为 y 则 $P(Y=0) = 1-2y$

$H(Y) = H(1-2y, 2y, 2y)$

$H(Y|X) = \sum_{k=0}^2 E(H(Y|X=k)) = 1.5p_1 + p_2$

$\therefore I(X; Y) = -p_1 + [1-2y \log_2 2y - (1-2y) \log_2 (1-2y)]$
 $\leq H_2(p_2)$ 当 $p_2 = 0.5$ 时最大. $I(X; Y) \leq 1 \text{ bit}$

(c)

$P = \begin{bmatrix} 1-\varepsilon & \varepsilon & 0 \\ 0 & 1-\varepsilon & \varepsilon \\ \varepsilon & 0 & 1-\varepsilon \end{bmatrix}$ 是一个强对称信道.
 $p(X) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ 即可达到容量.

$\therefore H(Y)_{\max} = \log_2 3$

$H(Y|X) = H(1-\varepsilon, \varepsilon, 0) = -\varepsilon \log_2 \varepsilon - (1-\varepsilon) \log_2 (1-\varepsilon) = H_2(\varepsilon)$

$\therefore C = \log_2 3 - H_2(\varepsilon)$

(d)

$P = \begin{bmatrix} 3/4 & 1/4 & 0 \\ 1/3 & 1/3 & 1/3 \\ 0 & 1/4 & 3/4 \end{bmatrix}$ 信道 0, 2 对称, 信道 1 是均匀分布.
 假设使其为 0, 即 $P(X=1) = 0$.
 $P(X=0) = P(X=2) = 1/2$

计算得到 $P(Y=0) = \frac{3}{8} \quad P(Y=1) = \frac{1}{4} \quad P(Y=2) = \frac{3}{8}$

$I(X=0; Y) = \frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} = 0.75 \text{ bit}$

$$I(X=2; Y) = \frac{1}{4} \log_2 \frac{\frac{1}{4}}{\frac{1}{4}} + \frac{3}{4} \log_2 \frac{\frac{3}{4}}{\frac{3}{8}} = 0.75 \text{ bit}$$

$$I(X=1; Y) = \frac{1}{3} \log_2 \frac{\frac{1}{3}}{\frac{1}{3/8}} + \frac{1}{3} \log_2 \frac{\frac{1}{3}}{\frac{1}{4}} + \frac{1}{3} \log_2 \frac{\frac{1}{3}}{\frac{1}{8}} \approx 0.05 \text{ bit} < 0.75 \text{ bit}$$

$$\therefore C = 0.75 \text{ bit}$$

(e.)

$$P = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

$$H(Y|X) = H(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0) = \log_2 3.$$

设 $p(x=i) = p_i$. $\therefore P(Y=0) = \frac{1}{3}p_0 + \frac{1}{3}p_2$ $P(Y=1) = \frac{1}{3}p_0 + \frac{1}{3}p_1$

$$P(Y=2) = \frac{1}{3}p_0 + \frac{1}{3}p_1 + \frac{1}{3}p_2 = \frac{1}{3}(p_1+p_2+p_3) = \frac{1}{3}$$

不妨设 $p(Y=0) = p$. 欲使 $H(Y)$ 最大. 则 $p(Y=0) = p(Y=1) = p$.

$$\text{又 } p + p + \frac{1}{3} = 1 \quad \therefore p = \frac{1}{3}$$

$$\therefore H(Y) = -\frac{1}{3} \log_2 \frac{1}{3} \cdot 3 = 3.$$

$$\therefore C = \log_2 3 - 3.$$

(f.)

$$P = \begin{bmatrix} 1-\varepsilon & \varepsilon \\ \delta & 1-\delta \end{bmatrix} \quad \text{设 } p(X=0) = p.$$

$$I(X=0; Y) = p \cdot (1-\varepsilon) \cdot \log_2 \frac{p(1-\varepsilon)}{p(p(1-\varepsilon) + (1-p)\delta)} + p \cdot \varepsilon \log_2 \frac{p \cdot \varepsilon}{p(p\varepsilon + (1-p)(1-\delta))}$$

$$I(X=1; Y) = (1-p) \delta \log_2 \frac{(1-p)\delta}{(1-p)(p(1-\varepsilon) + (1-p)\delta)} + (1-p)(1-\delta) \log_2 \frac{1-\delta}{p\varepsilon + (1-p)(1-\delta)}$$

$$\text{令 } I(X=0; Y) = I(X=1; Y)$$

为简便计算. 设 $\beta = p(Y=0)$.

得. $\log_2 \frac{\beta}{1-\beta} = \frac{H(\delta) - H(\varepsilon)}{1-\varepsilon-\delta} = A.$

$$\therefore \frac{\beta}{1-\beta} = 2^A. \quad \therefore \beta = \frac{2^A}{1+2^A}$$

$$\therefore C = \log_2 (1+2^A) - H(\varepsilon) - A(1-\varepsilon)$$

$$\text{其中 } A = \frac{H(\delta) - H(\varepsilon)}{1-\varepsilon-\delta}.$$

4.2.

观察到 1, 2 是对称的, 故令 $p(x=1) = p(x=2) = \frac{p}{2}$ $p(x=0) = 1-p$.

$$\therefore I(X=0; Y) = 1 \times \log_2 \frac{1}{(1-p)} = -\log_2(1-p).$$

$$I(X=1; Y) = (1-\varepsilon) \log_2 \frac{1-\varepsilon}{\frac{p}{2}} + \varepsilon \log_2 \frac{\varepsilon}{\frac{p}{2}}$$

$$\text{令 } I(X=0; Y) = I(X=1; Y)$$

$$\Rightarrow -\log_2(1-p) = (1-\varepsilon) \log_2 \frac{1-\varepsilon}{\frac{p}{2}} + \varepsilon \log_2 \frac{\varepsilon}{\frac{p}{2}}$$

$$\Rightarrow \frac{p}{1-p} = 2^{1-H(\varepsilon)} \quad \therefore C = \log_2(1+2^{1-H(\varepsilon)})$$

4.9

是强对称信道.

$\therefore H(Y|X) = H(p)$ 即使 $H(Y)$ 最大. 即平均分配

$$\therefore H(Y)_{\max} = \log_2 4 = 2 \text{ bit.}$$

$$\therefore I(X; Y) \leq 2 - H(p)$$

4.13.

$$C = \max_{p(x)} I(X; Y) = \max_{p(x)} [h(Y) - h(Y|X)]$$

$$h(Y|X) = h((X+Z) \bmod 2\pi | X) = h(Z \bmod 2\pi)$$

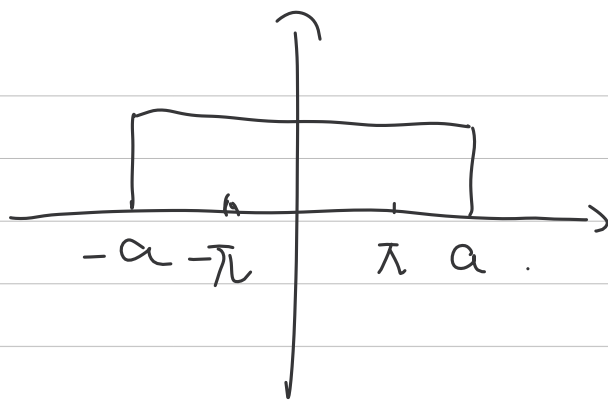
$$\text{令 } z' = Z \bmod 2\pi \quad \text{则 } h(Y|X) = h(z') \text{ 与 } X \text{ 无关}$$

$Y \in (-\pi, \pi)$ 当 Y 在 $(-\pi, \pi)$ 概率平均分布时, $H(Y)$ 最大
 $\therefore \max h(Y) = \log_2 2\pi$

$$(b) \text{ 当 } 0 \leq \pi \text{ 时 } z' = z. \quad \therefore h(z') = h(z) = \log_2 2a.$$

$$\therefore C = \log_2 2\pi - \log_2 2a = \log_2 \left(\frac{\pi}{a}\right)$$

$$(a) \text{ 当 } a > \pi \text{ 时 } \text{取 } n = \lfloor \frac{a}{\pi} \rfloor \quad A = 2a - n\pi.$$



$$h(z') = \begin{cases} \frac{n+1}{2a} & \text{在 } 2\Delta \text{ 范围内} \\ \frac{n}{2a} & \text{在 } 2\pi - 2\Delta \text{ 范围内} \end{cases}$$

$$h(z') = - \int_{-\pi}^{\pi} p(z') \log_2 p(z') dz'$$

$$= -2\Delta \frac{n+1}{2a} \log_2 \frac{n+1}{2a} - (2\pi - 2\Delta) \frac{n}{2a} \log_2 \frac{n}{2a}$$

$$\therefore C = \log_2 2\pi + \frac{\Delta(n+1)}{a} \log_2 \left(\frac{n+1}{2a} \right) + \frac{(2\pi - \Delta)n}{a} \log_2 \left(\frac{n}{2a} \right)$$

$$\text{其中 } n = \left\lfloor \frac{a}{\pi} \right\rfloor \quad \Delta = a - n\pi$$

4.14.

(1)

$$C = W \cdot \log_2 \left(1 + \frac{P}{WN_0} \right)$$

$$P = G_b \cdot C$$

$$\therefore C = W \log_2 \left(1 + \frac{G_b C}{WN_0} \right) \Rightarrow E_b = \frac{WN_0}{C} (2^{\frac{C}{W}} - 1)$$

$$\text{设 } x = \frac{C}{W} \quad \therefore E_b = N_0 \cdot \frac{2^x - 1}{x} \quad E_{b \min} = \lim_{x \rightarrow 0} N_0 \cdot \frac{2^x - 1}{x} = N_0 \ln 2$$

此时 $W \rightarrow \infty$

$$\therefore E_{b \min} = 0.693 N_0$$

(2)

$$\text{设 } y = \frac{P}{WN_0}$$

$$\therefore P = y W N_0 \quad y > 4000$$

$$E_b = \frac{P}{C} = \frac{P}{W \log_2 \left(1 + \frac{P}{WN_0} \right)} = \frac{y W N_0}{W \log_2 \left(1 + \frac{y}{4000} \right)} = N_0 \frac{y}{\log_2 (1+y)}$$

$$E_b > N_0 \frac{4000}{\log_2 (1+4000)}$$

$$\therefore \beta = \frac{N_0 \frac{4000}{\log_2 (4001)}}{N_0 \ln 2} \approx 482$$

